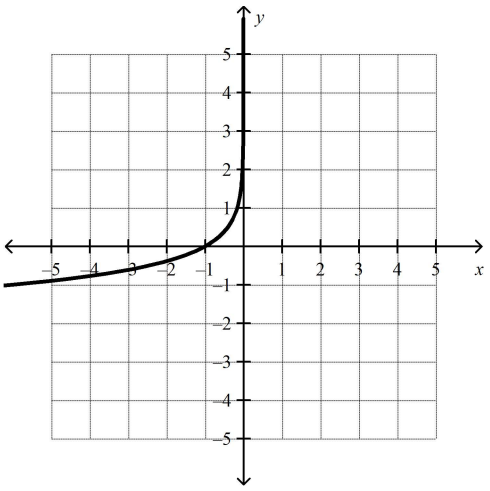
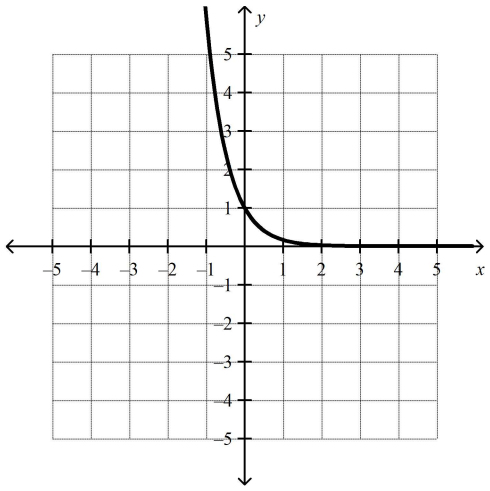
MULTIPLE CHOICE

1. Which graph represents the inverse of $y = \left(\frac{1}{6}\right)^x$?

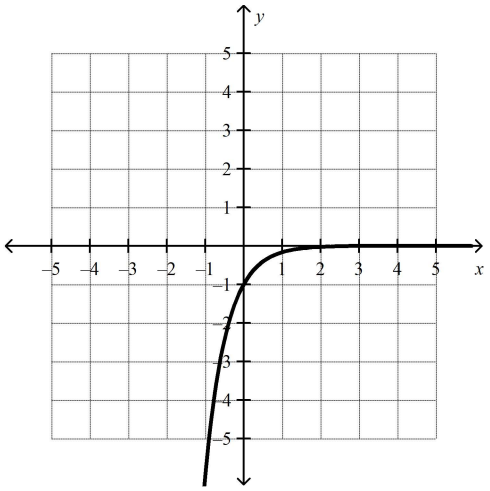
A.



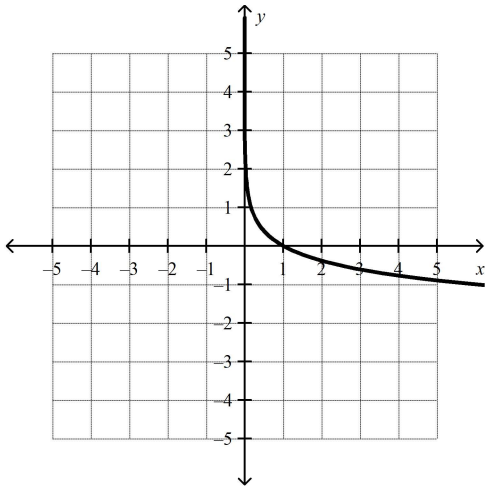
C.



B.



D.



ANS: D DIF: Easy OBJ: Section 8.1 NAT: RF7
TOP: Understanding Logarithms KEY: graph | inverse

2. Another way of writing $5^5 = 3125$ is

A. $\log_5 5 = 3125$

B. $\log_5 3125 = 5$

C. $\log_{3125} 5 = 5$

D. $\log_5 5 = 3125$

ANS: B DIF: Easy OBJ: Section 8.1 NAT: RF7
TOP: Understanding Logarithms KEY: logarithm | exponential function
NOT: Draft

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3. Which of the following represents $a = \log_7 343$?

A. $7^{343} = a$
B. $7^a = 343$
C. $a^7 = 343$
D. $a^{343} = 7$

ANS: B DIF: Easy OBJ: Section 8.1 NAT: RF7
TOP: Understanding Logarithms KEY: logarithm | exponential function
NOT: Draft

4. Another way of writing $7^{-3} = \frac{1}{343}$ is

A. $\log_7\left(\frac{1}{3}\right) = -343$
B. $\log_3\left(-\frac{1}{7}\right) = 343$
C. $\log_7\left(\frac{1}{343}\right) = -3$
D. $\log_7\left(-\frac{1}{343}\right) = -3$

ANS: C DIF: Easy OBJ: Section 8.1 NAT: RF7
TOP: Understanding Logarithms KEY: logarithm | exponential function
NOT: Draft

5. Evaluate $\log_4 65\,536$.

A. 4096
B. 5.33
C. 0.13
D. 8

ANS: D DIF: Average OBJ: Section 8.1 NAT: RF7
TOP: Understanding Logarithms KEY: logarithm

6. The range of the function $f(x) = -8\log_5[10(x+8)] + 9$ is

A. $\{y | y \in \mathrm{R}\}$
B. $\{y | y \geq 8, y \in \mathrm{R}\}$
C. $\{y | y \geq 9, y \in \mathrm{R}\}$
D. $\{y | y \leq 9, y \in \mathrm{R}\}$

ANS: A DIF: Easy OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions KEY: range | logarithmic functions

7. The domain of the function $f(x) = 8\log_6[8(x+8)] + 7$ is

A. $\{x | x < 7, x \in \mathrm{R}\}$
B. $\{x | x < -8, x \in \mathrm{R}\}$
C. $\{x | x > 7, x \in \mathrm{R}\}$
D. $\{x | x > -8, x \in \mathrm{R}\}$

ANS: D DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: domain | logarithmic functions

8. Compared to the graph of the base function $y = \log_{10} x$, the graph of the function $y = \log_{10} x + 4$ is translated

A. 4 units to the left
B. 4 units down
C. 4 units up
D. 4 units to the right

ANS: C DIF: Easy OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: vertical translation | transformation

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9. Compared to the graph of the base function $y = \log_7 x$, the graph of the function $y = \log_7(x-6)+7$ is translated

A. 6 units to the right and 7 units up
B. 7 units to the left and 6 units down
C. 7 units to the right and 6 units up
D. 6 units to the left and 7 units down

ANS: A DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | vertical translation

10. Compared to the graph of the base function $f(x) = \log_9 x$, the graph of the function $g(x) = 3\log_9[3(x+6)] - 10$ is

A. translated down 10 units and left 6 units, horizontally stretched by a factor of $\frac{1}{3}$, reflected in the y-axis, vertically stretched by a factor of 3, and reflected in the x-axis
B. translated down 10 units and right 6 units, horizontally stretched by a factor of $\frac{1}{3}$, reflected in the y-axis, vertically stretched by a factor of 3, and not reflected in the x-axis
C. translated down 10 units and left 6 units, horizontally stretched by a factor of $\frac{1}{3}$, not reflected in the y-axis, vertically stretched by a factor of 3, and not reflected in the x-axis
D. translated up 10 units and left 6 units, horizontally stretched by a factor of $\frac{1}{3}$, not reflected in the y-axis, vertically stretched by a factor of 3, and not reflected in the x-axis

ANS: C DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal stretch | vertical translation | reflection

11. Which function represents a vertical translation of 7 units down, a horizontal translation of 8 units right, a horizontal stretch by a factor of $\frac{1}{6}$, no reflection in the y-axis, a vertical stretch by a factor of 6, and no reflection in the x-axis, when compared to the base function $f(x) = \log_9 x$.

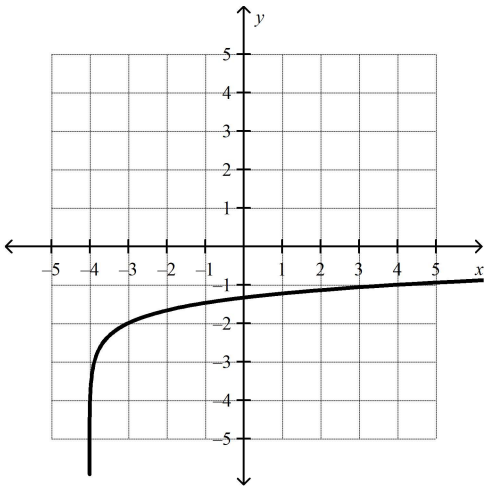
A. $g(x) = 6\log_9\left[\frac{1}{6}(x-8)\right] - 7$
B. $g(x) = \frac{1}{6}\log_9\left[6(x-8)\right] - 7$
C. $g(x) = 6\log_9\left[6(x-7)\right] - 8$
D. $g(x) = 6\log_9\left[6(x-8)\right] - 7$

ANS: D DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal stretch | vertical translation | reflection

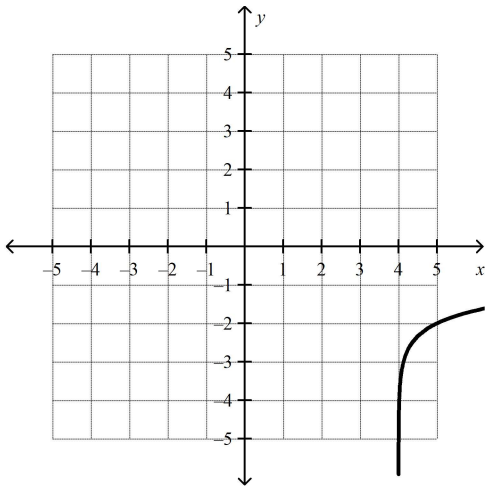
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12. Which graph represents the function $y = \log_8(x-2) - 4$?

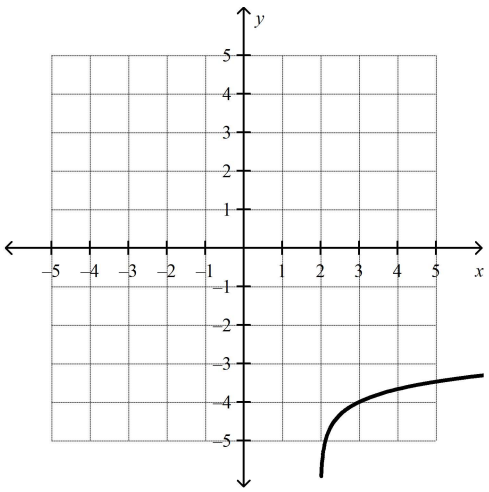
A.



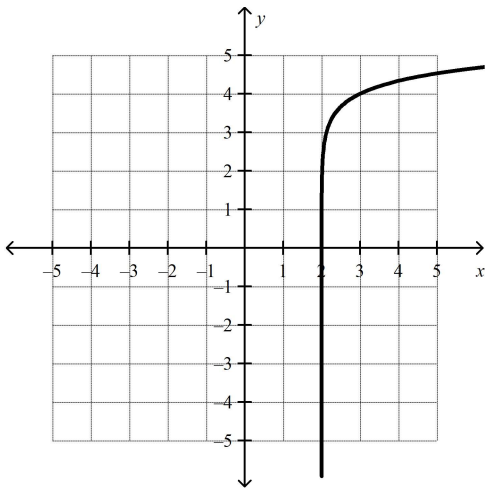
C.



B.



D.



ANS: B DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | vertical translation

13. What is the equation for the asymptote of the function $f(x) = -\log_7[-5(x+2)] - 3$?

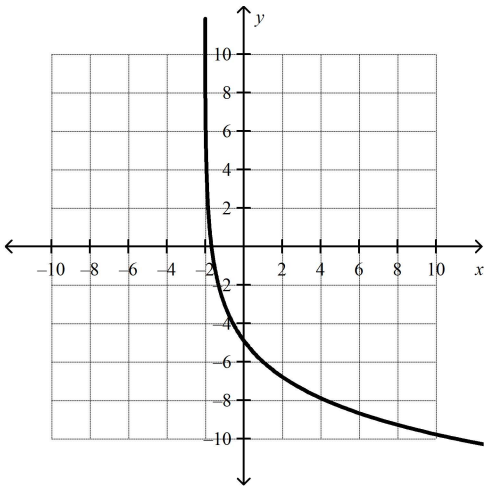
A. $x = 2$
B. $x = -3$
C. $x = -5$
D. $x = -2$

ANS: D DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | asymptote

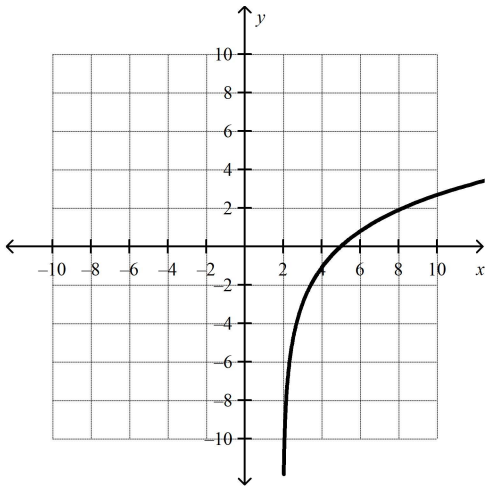
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14. Which graph represents the function $y = -3\log_3[(x-2)] - 3$?

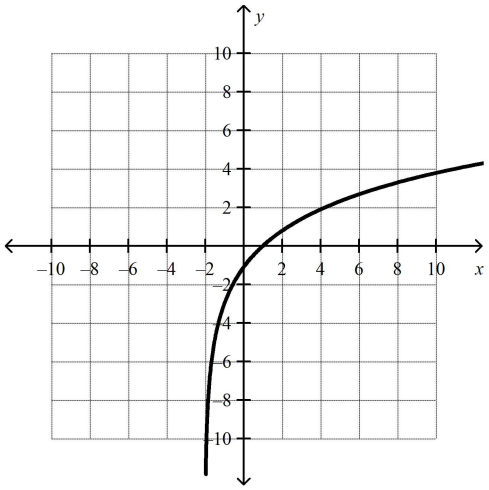
A.



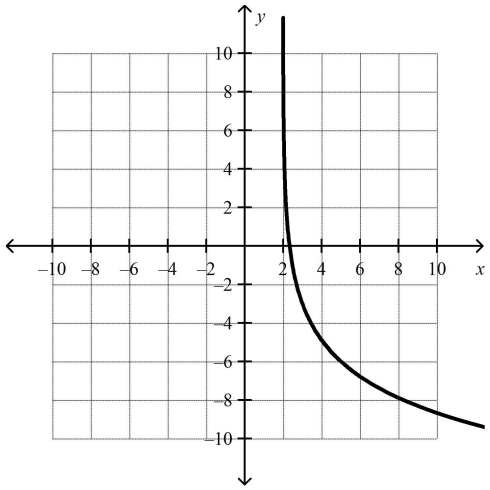
C.



B.



D.

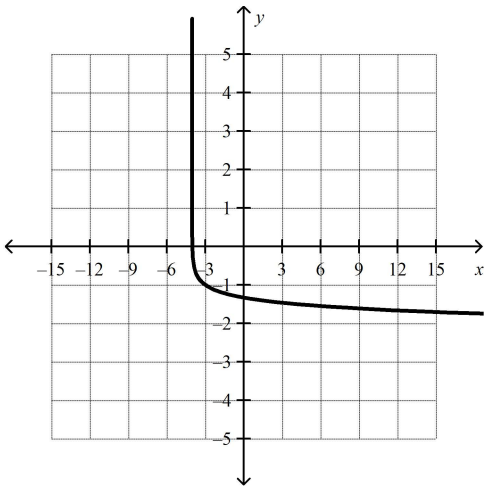


ANS: D DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | vertical translation | vertical stretch | horizontal stretch

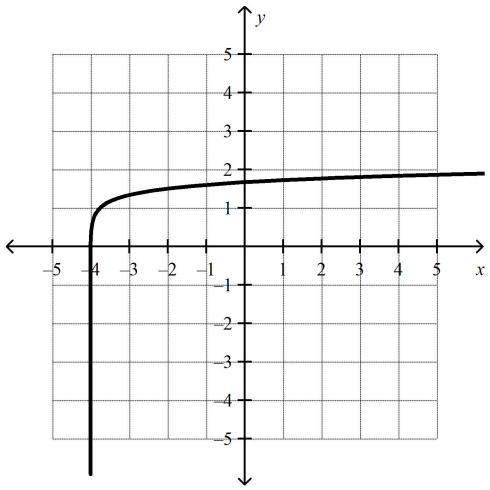
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15. Which graph represents the function $y = \frac{1}{3}\log_4(x-3)+4$?

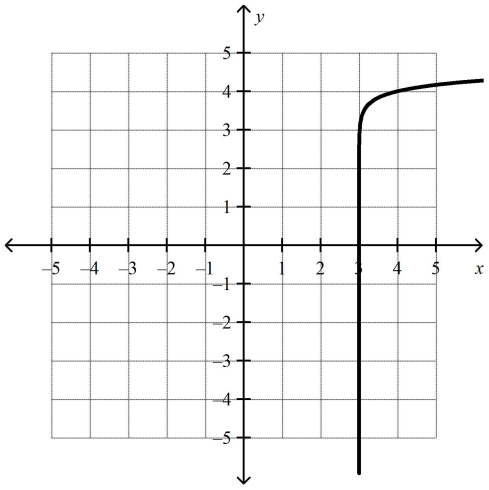
A.



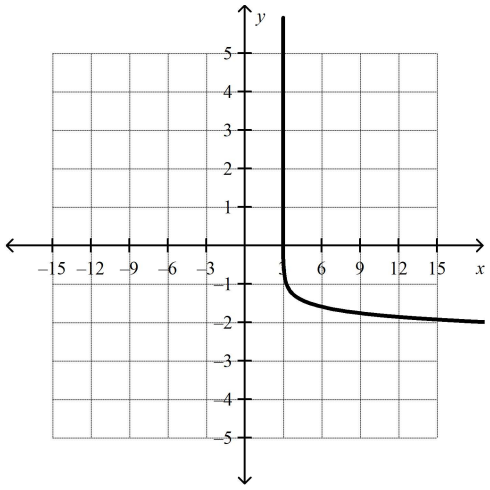
C.



B.



D.



ANS: B DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | vertical translation | vertical stretch

16. Which of the following is equivalent to the expression $\log_4 s + 7\log_4 v + \log_4 z$?

A. $\log_4 sv^7 z$
B. $7\log_4 svz$
C. $\log_4 7svz$
D. $\log_4 sz + \log_{28} v$

ANS: A DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: product law | laws of logarithms

17. If $\log 3 = s$, $\log 5 = v$, and $\log 7 = z$, an algebraic expression in terms of s, v, and z for $\log \frac{5}{441}$ is

A. $v - 2s + 2z$
B. $v - 2(s + z)$
C. $v - 2(s - z)$
D. $v - 2s + z$

ANS: B DIF: Average OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: product law | laws of logarithms | quotient law

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18. Which if the following is equivalent to the expression $\log_4 sw^{10}y$?

A. $\log_4 s + 10\log_4 w + \log_4 y$

B. $10\log_4 s - 10\log_4 w + \log_4 y$

C. $\log_4 s + \log_4 w + 10\log_4 y$

D. $10\log_4 s + \log_4 w + \log_4 y$

ANS: A DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: product law | laws of logarithms

19. Evaluate the expression $6\log_8 25 - (\log_8 3 + \log_8 8)$ to the nearest hundredth.

A. 7.76

B. 9.76

C. 10.82

D. 7.01

ANS: A DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: product law | laws of logarithms | quotient law

20. The pH scale is used to measure the acidity or alkalinity of a solution. pH is defined as $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydronium ions, measured in moles per litre. Determine the pH of a solution with a concentration of $[\text{H}^+] = 4.3 \times 10^{-6}$. Round your answer to two decimal places.

A. 6.00

B. 5.37

C. 0.78

D. 3.52

ANS: B DIF: Easy OBJ: Section 8.1 | Section 8.3
NAT: RF7 | RF9 TOP: Understanding Logarithms | Laws of Logarithms
KEY: logarithmic equation

21. Solve $\log_3 x = \log_3 2 + \log_3 3$.

A. $x = 6$

B. $x = \frac{3}{2}$

C. $x = \frac{2}{3}$

D. $x = 729$

ANS: A DIF: Average OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations KEY: logarithmic equation

22. Solve $10^{2x-5} = 7^{x+4}$. Round your answer to two decimal places.

A. 3.06

B. 7.26

C. 2.95

D. –1.40

ANS: B DIF: Average OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: logarithmic equation | exponential equation

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23. A graduate student determines that the relationship between the length, s, in metres, of the skull and the body mass, m, in kilograms, of a particular species can be expressed using the equation $3.6045\log s = \log m - 3.4425$. Determine the body mass of an animal with a skull size of 0.56 m. Round your answer to the nearest kilogram.

A. 547 kg
C. 343 kg
B. 376 kg
D. 313 kg

ANS: C DIF: Average OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: exponential equation | logarithmic equation

24. Solve $\log(3x + 15) = 1 + \log(x + 3)$ to the nearest hundredth.

A. –6.43
C. 1.15
B. 3.46
D. –2.14

ANS: D DIF: Average OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations KEY: logarithmic equation

25. Solve $8^x = 486$. Round your answer to two decimal places.

A. 3.59
C. 1.78
B. 2.97
D. 0.34

ANS: B DIF: Easy OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations KEY: exponential equation

26. The eye size of many vertebrates is related to body mass by the logarithmic equation $\log E = \log 10.61 + 0.1964\log m$, where E is the eye axial length, in millimetres, and m is the body mass, in kilograms. Predict the mass of a vertebrate with an eye axial length of 43 mm. Round your answer to the nearest hundredth of a kilogram.

A. 2.66
C. 1242.98
B. 868.60
D. 1.32

ANS: C DIF: Difficult OBJ: Section 8.1 | Section 8.4
NAT: RF10 | RF7 TOP: Understanding Logarithms | Logarithmic and Exponential Equations
KEY: exponential equation | logarithmic equation

COMPLETION

1. The function $y = 9\log_6[10(x - 6)] + 8$ written in exponential form is _____ .

ANS: $x = \dfrac{6^{\left(\frac{y}{9} - 8/9\right)}}{10} + 6$

DIF: Average OBJ: Section 8.1 NAT: RF7 | RF10 TOP: Understanding Logarithms
KEY: logarithmic functions | exponential function NOT: draft

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2. The variable k in the function $f(x) = a\log_c[b(x-h)] + k$ represents _____.

ANS: a vertical translation

DIF: Easy OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: logarithmic functions | transformation

3. The graph of $g(x) = \log_c(x-6)$ is the graph of $f(x)$ translated _____.

ANS: right 6 units

DIF: Easy OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: horizontal translation | vertical translation

4. The domain of the function $f(x) = 5\log_8[-9(x-9)] + 4$ is _____.

ANS: $\{x|x < 9, x \in \mathrm{R}\}$

DIF: Easy OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: domain | horizontal translation NOT: Draft

5. The equation $\log_c M^P = P\log_c M$ is an example of the _____ law of logarithms.

ANS: power

DIF: Easy OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: laws of logarithms

MATCHING

Match the single logarithm in simplest form with the correct equivalent expression.

A. $\log_7 s - \log_7 u + 3\log_7 x$
B. $8\log_7 s - \log_7 u - 3\log_7 x$
C. $8/3\log_7 u - 8\log_7 s - 3\log_7 x$
D. $8/3\log_7 u - 8\log_7 s + 3\log_7 x$
E. $8\log_7 s - 8/3\log_7 u + 3\log_7 x$
F. $8\log_7 s + 8/3\log_7 u + 3\log_7 x$

1. $\log_7 \dfrac{u^{8/3}}{s^8 x^3}$

2. $\log_7 \dfrac{sx^3}{u}$

3. $\log_7 \dfrac{u^{8/3}x^3}{s^8}$

4. $\log_7 \dfrac{s^8 x^3}{u^{8/3}}$

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5. $\log_7 \frac{s^8}{ux^3}$

1. ANS: C DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: power law | product law | quotient law
2. ANS: A DIF: Easy OBJ: Section 8.3 NAT: RF10
TOP: Laws of Logarithms KEY: power law | product law | quotient law
3. ANS: D DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: power law | product law | quotient law
4. ANS: E DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: power law | product law | quotient law
5. ANS: B DIF: Easy OBJ: Section 8.3 NAT: RF9
TOP: Laws of Logarithms KEY: power law | product law | quotient law

SHORT ANSWER

1. Evaluate $\log_2 \sqrt[4]{32}$.

ANS:
$\frac{5}{4}$

DIF: Average OBJ: Section 8.1 NAT: RF7 TOP: Understanding Logarithms
KEY: logarithm NOT: Draft

2. Evaluate $\log_2 64 + \log_3 27 \times \log_4 \frac{1}{256}$.

ANS:

$$\log_2 64 + \log_3 27 \times \log_4 \left(\frac{1}{256}\right) = 6 + 3(-4)$$
$$= 6 - 12$$
$$= -6$$

DIF: Difficult OBJ: Section 8.1 NAT: RF7 TOP: Understanding Logarithms
KEY: logarithm | exponential function NOT: Draft

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3. If $\log_4(m-n)=0$ and $\log_4(m+n)=2$, determine the values of m and n.

ANS:
Solve the system of equations:
$m-n=4^0$ and $m+n=4^2$

$m-n=1$

$m+n=16$
Add these equations to find m:
$2m=17$

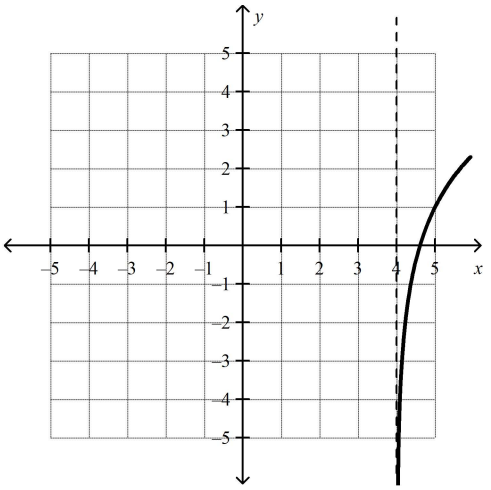
$m=8.5$
Subtract the first equation from the second to find n:
$2n=15$

$n=7.5$

DIF: Difficult OBJ: Section 8.1 NAT: RF7 TOP: Understanding Logarithms
KEY: logarithm | simultaneous equations

4. Sketch the graph of the function $y=2\log(x-4)+1$.

ANS:

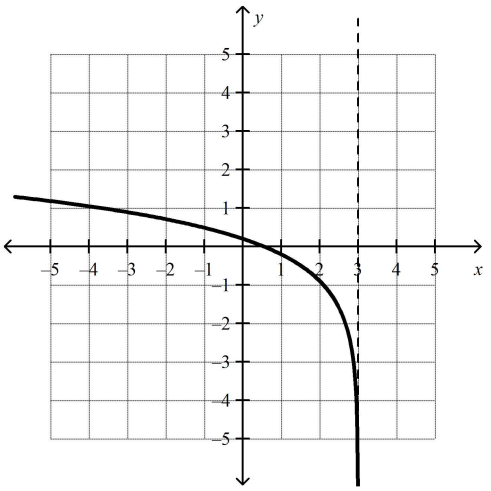


DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: transformations | vertical translation | horizontal translation

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5. Sketch the graph of the function $y = \log(-3x + 9) - 2$.

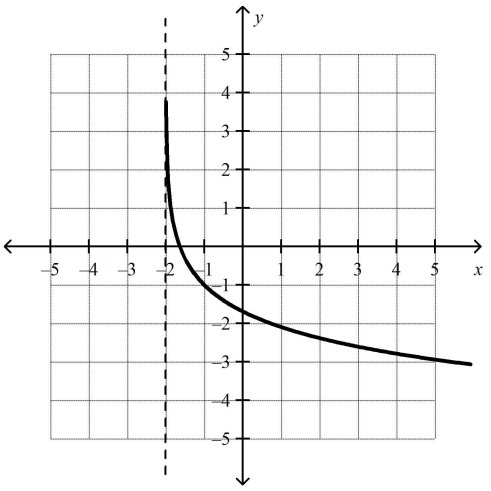
ANS:



DIF: Difficult OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: transformation | vertical translation | horizontal translation | horizontal stretch

6. Graph the function $f(x) = -\log(x + 2) - 1$. Identify the domain, the range, and the equation of the vertical asymptote.

ANS:



domain: $\{x | x > -2, x \in \mathrm{R}\}$
range: $\{y | y \in \mathrm{R}\}$
equation of vertical asymptote: $x = -2$

DIF: Difficult OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: transformation | vertical translation | asymptote | graph

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7. A certain type of exponential growth can be described by the equation $N = N_0 10^{kt}$, where N_0 is the initial amount; k is the doubling time, in years; and N is the amount after time, t, in years, has passed. Suppose that the population of a small town doubles every 25 years. How long does it take to triple, to the nearest tenth of a year?

ANS:
First solve for k.

$$N = N_0 10^{kt}$$

$$2N_0 = N_0 10^{k(25)}$$

$$2 = 10^{25k}$$

$$\log 2 = 25k \log 10$$

$$k = \frac{\log 2}{25}$$

Solve for t when $N = 3N_0$.

$$3N_0 = N_0 10^{\left(\frac{\log 2}{25}\right)t}$$

$$3 = 10^{\left(\frac{\log 2}{25}\right)t}$$

$$\log 3 = \left(\frac{\log 2}{25}\right)t$$

$$t = \left(\frac{25}{\log 2}\right)\log 3$$

$$t \approx 39.62$$

It will take 39.6 years for the population to triple.

DIF: Difficult + OBJ: Section 8.3 | Section 8.4 NAT: RF9
TOP: Laws of Logarithms | Logarithmic and Exponential Equations
KEY: power law | law of logarithms | logarithmic equation | exponential equation

8. Given $\log_2 7 \approx 2.8074$, find the value of $\log_2 14$.

ANS:

$$\log_2 14 = \log_2 (2 \times 7)$$

$$= \log_2 2 + \log_2 7$$

$$\approx 1 + 2.8074$$

$$= 3.8074$$

DIF: Difficult OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: power law | laws of logarithms

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9. Solve the equation $6^{3x+1} = 2^{2x-3}$. Leave your answer in exact form.

ANS:

$$6^{3x+1} = 2^{2x-3}$$

$$\log(6^{3x+1}) = \log(2^{2x-3})$$

$$(3x+1)\log 6 = (2x-3)\log 2$$

$$3x\log 6 + \log 6 = 2x\log 2 - 3\log 2$$

$$x(3\log 6 - 2\log 2) = -3\log 2 - \log 6$$

$$x = \frac{-(3\log 2 + \log 6)}{3\log 6 - 2\log 2}$$

DIF: Average OBJ: Section 8.3 | Section 8.4 NAT: RF9
TOP: Laws of Logarithms | Logarithmic and Exponential Equations
KEY: exponential equation | laws of logarithms

10. Solve for x.
$2\log_4(x+4) - \log_4(x+12) = 1$

ANS:
$2\log_4(x+4) - \log_4(x+12) = 1$

$$\log_4(x+4)^2 - \log_4(x+12) = 1$$

$$\log_4 \frac{(x+4)^2}{(x+12)} = \log_4 4^1$$

$$\frac{(x+4)^2}{(x+12)} = 4$$

$$(x+4)^2 = 4x + 48$$

$$x^2 + 8x + 16 = 4x + 48$$

$$x^2 + 4x - 32 = 0$$

$$(x+8)(x-4) = 0$$

$$x = -8, x = 4$$

Since $x = -8$ is an extraneous root, the solution is $x = 4$.

DIF: Average OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: logarithmic equation | laws of logarithms NOT: Draft

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PROBLEM

1. Dye spilled in a bucket of water mixes with the water at a rate defined by $R = -236\log\left(\frac{V}{80}\right)$, where R is the intensity of the dye and V is the volume of dye spilled, in millilitres. What is the intensity of the mixture if 50 mL of dye was spilled? Round your answer to two decimal places.

ANS:

Substitute $V = 50$ into the formula $R = -236\log\left(\frac{V}{80}\right)$ and solve for R.

$R \doteq 48.17$

DIF: Average OBJ: Section 8.1 NAT: RF7 TOP: Understanding Logarithms
KEY: logarithm NOT: Draft

2. The magnitude of an earthquake is defined as $M = \log\left(\frac{A}{A_0}\right)$, where A is the amplitude of the ground motion and A_0 is the amplitude corrected for the distance from the actual earthquake that would be expected for a "standard earthquake." On March 2, 2012, an earthquake with an amplitude $10^{5.1}$ times A_0 was recorded in Norman Wells, Northwest Territories.
a) What was the earthquake's magnitude on the Richter scale?
b) How does the earthquake in Norman Wells compare to the earthquake off Vancouver Island in 1946 that measured 7.3 on the Richter scale?

ANS:

a) $M = \log\left(\frac{10^{5.1}A_0}{A_0}\right)$

$M = \log 10^{5.1}$

$M = 5.1$

The earthquake in Norman Wells measured 5.1 on the Richter scale.

b) $\frac{\text{Vancouver Island amplitude}}{\text{Norman Wells amplitude}} = \frac{10^{7.3}A_0}{10^{5.1}A_0}$

$= \frac{10^{7.3}}{10^{5.1}}$

≈ 158

The earthquake off Vancouver Island was about 158 times as strong as the earthquake off Norman Wells.

DIF: Average OBJ: Section 8.1 NAT: RF7 TOP: Understanding Logarithms
KEY: logarithm NOT: Draft

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3. List the steps and explain the effect of each transformation to graph the function $y = -3\log\left[-2(x-1)\right] + 4$.

ANS:
1. vertically stretched by a factor of 3
2. horizontally compressed by a factor of $\frac{1}{2}$,
3. reflected in both the x- and y-axes
4. translated 1 unit right and 4 units up

DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: transformations | logarithmic functions

4. Given $\log 7 \approx 0.8451$ and $\log 2 \approx 0.3010$, find the value of $\log 28$.

ANS:

$$\begin{aligned}\log 28 &= \log(7 \times 2^2) \\ &= \log 7 + 2\log 2 \\ &\approx 0.8451 + 2(0.3010) \\ &= 1.4471\end{aligned}$$

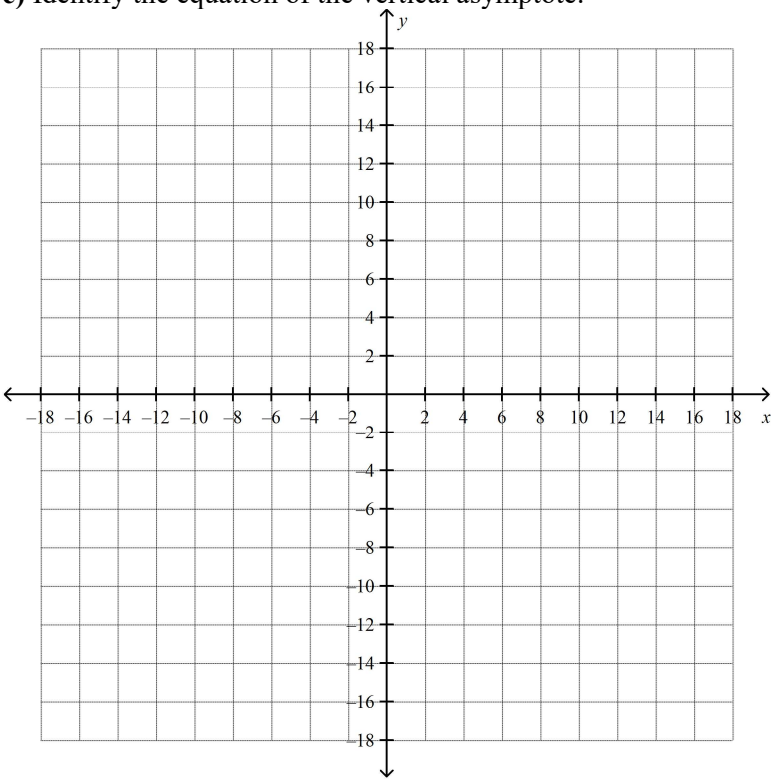
DIF: Difficult OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: power law of logarithms

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5. **a)** Sketch the graph of the function $f(x) = 3\log\left[2(x-1)\right] + 4$. Identify the key features.

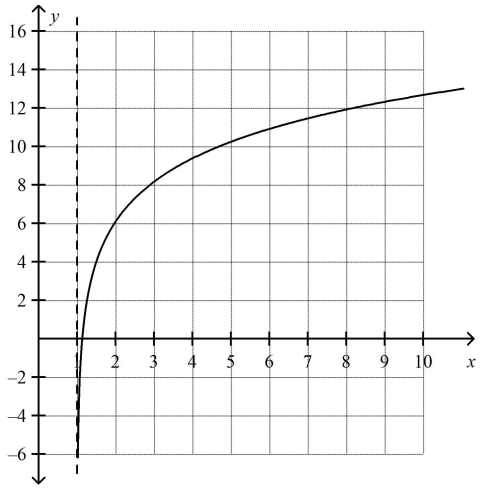
b) Identify the domain and range.

c) Identify the equation of the vertical asymptote.



ANS:

a)



b) domain: $\{x | x > 1, x \in \mathrm{R}\}$; range: $\{y | y \in \mathrm{R}\}$

c) $x = 1$

DIF: Difficult OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions
KEY: transformation | domain | range | asymptote

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6. The intensity level, β, in decibels, of sound is defined as $\beta = 10\log\left(\frac{I}{I_0}\right)$, where I is the intensity of the sound in watts per square metre.The ratio of the intensity of sound pollution measured at a small airport runway versus that of the local highway is 6420.4. If the sound level on the local highway is 91 dB, determine the sound level on the runway, to the nearest decibel.

ANS:
Let β_1 represent the sound intensity at the airport and β_2 represent the sound intensity on the highway.

$$\beta_1 - \beta_2 = 10\log\left(\frac{I_1}{I_0}\right) - 10\log\left(\frac{I_2}{I_0}\right)$$

$$\beta_1 - \beta_2 = 10\log\left(\frac{\frac{I_1}{I_0}}{\frac{I_2}{I_0}}\right)$$

$$\beta_1 - \beta_2 = 10\log\left(\frac{I_1}{I_2}\right)$$

$$\beta_1 - 91 = 10\log(6420.4)$$

$$\beta_1 = 10\log 6420.4 + 91$$

$$\beta_1 \approx 129$$

The sound level on the runway is 129 dB.

DIF: Difficult OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: laws of logarithms

7. Find the difference in pH levels of a swimming pool with a hydronium ion concentration of 4.4×10^{-8} mol/L and a swimming pool with a hydronium ion concentration of 5.7×10^{-8} mol/L, to the nearest hundredth.

ANS:
Find the pH of each pool individually, and then subtract.
Pool 1: $\text{pH} = -\log(4.4 \times 10^{-8})$

$$\approx 7.36$$

Pool 2: $\text{pH} = -\log(5.7 \times 10^{-8})$

$$\approx 7.24$$

The difference in pH is 0.12.

DIF: Difficult OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: laws of logarithms

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8. The stellar magnitude scale compares the brightness of stars using the equation $m_2 - m_1 = \log\left(\frac{b_1}{b_2}\right)$, where m_1 and m_2 are the apparent magnitudes (how bright the stars appear in the sky) of the two stars being compared, and b_1 and b_2 are their brightness (how much light they emit).

a) The brightest appearing star in our sky, Sirius, has an apparent magnitude of –1.5. How much brighter does Sirius appear than Betelgeuse, whose apparent magnitude is 0.12? Round your answer to the nearest whole number.

b) The Sun appears about 1.3×10^{10} times as bright in the sky as does Sirius. What is the apparent magnitude of the Sun, to the nearest tenth?

ANS:

Let m_1 represent the apparent magnitude of Sirius, b_1 represent the brightness of Sirius, m_2 represent the apparent magnitude of the Sun, and b_2 represent the brightness of the Sun.

a) $m_2 - m_1 = \log\left(\frac{b_1}{b_2}\right)$

$$0.12 + 1.5 = \log\left(\frac{b_1}{b_2}\right)$$

$$1.62 = \log\left(\frac{b_1}{b_2}\right)$$

$$10^{1.62} = \frac{b_1}{b_2}$$

$$\frac{b_1}{b_2} \approx 41.69$$

Sirius is approximately 42 times as bright as the Sun.

b) $m_1 - m_2 = \log\left(\frac{b_2}{b_1}\right)$

$$-1.5 - m_2 = \log(1.3\times10^{10})$$

$$m_2 \approx -1.5 - 10.11$$

$$m_2 \approx -11.61$$

The apparent magnitude of the Sun is –11.6.

DIF: Difficult OBJ: Section 8.1 | Section 8.3 | Section 8.4
NAT: RF7 | RF9
TOP: Understanding Logarithms | Laws of Logarithms | Logarithmic and Exponential Equations
KEY: logarithmic equation | exponential function

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9. Prove that $\log a + \log a^2 + \log a^3 - \log a^6 = \log 1$.

ANS:

$\mathbf{L.S.} = \log a + \log a^2 + \log a^3 - \log a^6$

$= \log a + 2\log a + 3\log a - 6\log a$

$= (1 + 2 + 3 - 6)\log a$

$= 0 \log a$

$= 0$

$\mathbf{R.S.} = \log 1$

$= 0$

L.S. = R.S.

Thus, $\log 5 + \log 5^2 + \log 5^3 - \log 5^6 = \log 1$.

DIF: Average OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: power law | laws of logarithms

10. Show that $\dfrac{1}{\log_a b} = \log_b a$.

ANS:

$\mathbf{L.S.} = \dfrac{1}{\log_a b}$

$= 1 \div \log_a b$

$= 1 \div \dfrac{\log b}{\log a}$

$= 1 \times \dfrac{\log a}{\log b}$

$= \log_b a$

$\mathbf{R.S.} = \log_b a$

L.S. = R.S.

Thus, $\dfrac{1}{\log_a b} = \log_b a$.

DIF: Difficult + OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: change of base formula | laws of logarithms

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11. Show that $3\log\sqrt{x}+2\log x-\frac{1}{2}\log x=3\log x$.

ANS:

$\mathbf{L.S.}=3\log\sqrt{x}+2\log x-\log\sqrt{x}$ $\qquad$ $\mathbf{R.S.}=3\log x$

$$=\log x^{\frac{3}{2}}+\log x^2-\log x^{\frac{1}{2}}$$

$$=\log\left[\frac{x^{\frac{3}{2}}x^2}{x^{\frac{1}{2}}}\right]$$

$$=\log x^{\frac{3}{2}+2-\frac{1}{2}}$$

$$=\log x^3$$

$$=3\log x$$

L.S. = **R.S.**

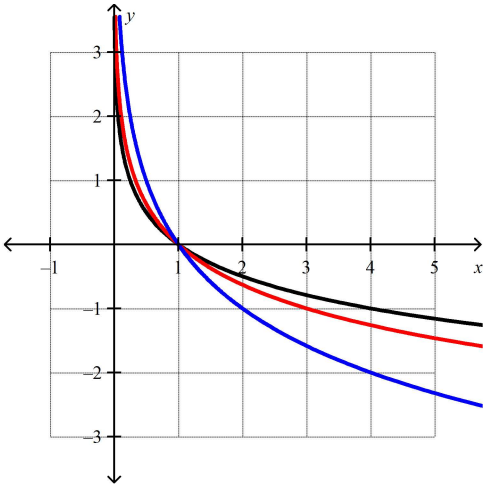
Thus, $3\log\sqrt{x}+2\log x-\frac{1}{2}\log x=3\log x$.

DIF: Average OBJ: Section 8.3 NAT: RF9 TOP: Laws of Logarithms
KEY: laws of logarithms | power law | quotient law | product law

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12. Sketch the graphs of $y = \log_{\frac{1}{2}} x$, $y = \log_{\frac{1}{3}} x$, and $y = \log_{\frac{1}{4}} x$, and then describe the pattern in terms of n for $y = \log_{\frac{1}{n}} x$.

ANS:



$y = \log_{\frac{1}{2}} x$: blue

$y = \log_{\frac{1}{3}} x$: red

$y = \log_{\frac{1}{4}} x$: black

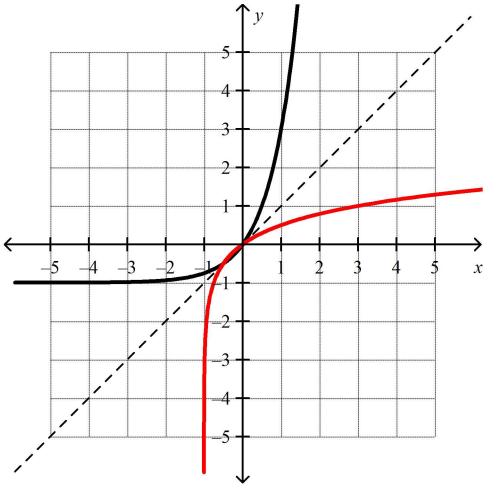
The point (1, 0) is the x-intercept for all of the graphs. As n increases, the negative part of the graph of $y = \log_{\frac{1}{n}} x$ approaches the x-axis, getting flatter and flatter. The positive part gets steeper and steeper.

DIF: Difficult + OBJ: Section 8.2 NAT: RF10
TOP: Transformations of Logarithmic Functions
KEY: graph | transformations of logarithmic functions

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13. Sketch a graph of the function $f(x) = 2^{2x} - 1$ and its inverse $f^{-1}(x)$. Compare the graph of $f^{-1}(x)$ to the graph of the function $g(x) = \frac{1}{2}\log_2(x+1)$. What conclusions can you make?

ANS:



$f^{-1}(x) = g(x)$ for the functions.

DIF: Difficult OBJ: Section 8.2 | Section 8.4 NAT: RF10
TOP: Transformations of Logarithmic Functions | Logarithmic and Exponential Equations
KEY: exponential function | transformation | exponential function

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14. A 200-g sample of a radioactive substance is placed in a chamber to be tested. After 3 h, 140 g of the sample remains.
a) Determine the half-life of this substance, to the nearest hundredth of an hour.
b) Graph the amount of the substance remaining as a function of time.

ANS:
a) Let h represent the half-life of the substance.

$$140 = 200\left(\frac{1}{2}\right)^{\frac{3}{h}}$$

$$0.7 = 0.5^{\frac{3}{h}}$$

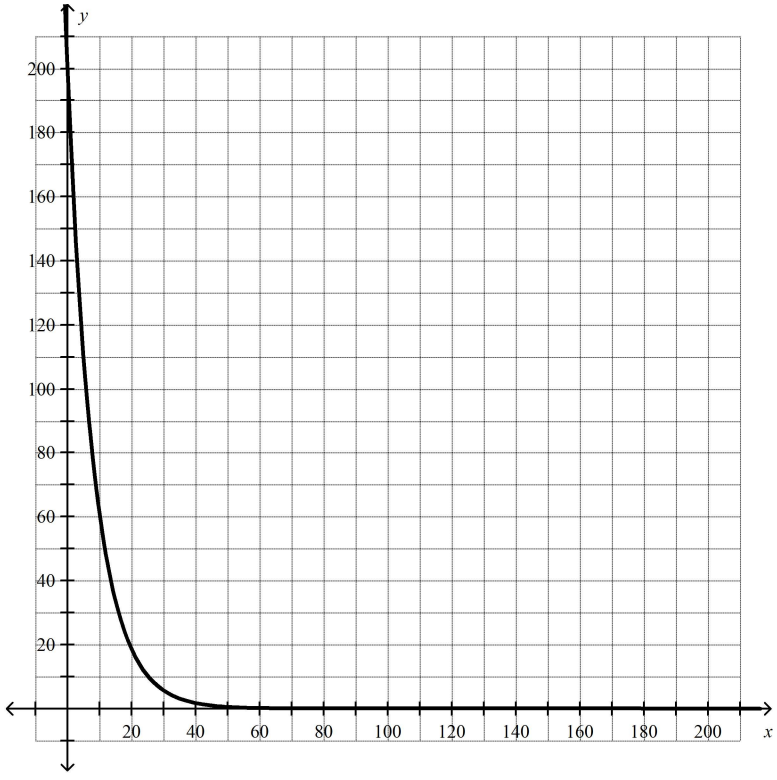
$$\log 0.7 = \frac{3}{h}\log 0.5$$

$$h = \frac{3\log 0.5}{\log 0.7}$$

$$h \approx 5.83$$

The half-life is 5.83 h.

b) Graph $y = 200\left(\frac{1}{2}\right)^{\frac{x}{5.83}}$.



DIF: Difficult OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: exponential equation | logarithmic equation

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15. Prove that $\log_{q^5} p^5 = \log_q p$.

ANS:

$\mathbf{L.S.} = \log_{q^5} p^5$ $\qquad$ $\mathbf{R.S.} = \log_q p$

$$= \frac{\log_q p^5}{\log_q q^5}$$

$$= \frac{5\log_q p}{5}$$

$$= \log_q p$$

L.S. = R.S.

Thus, $\log_{q^5} p^5 = \log_q p$.

DIF: Difficult + OBJ: Section 8.3 NAT: RF10 TOP: Laws of Logarithms
KEY: laws of logarithms

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16. **a)** Graph the function $y = \log_3 x$.

b) Graph the following functions on the same graph:

$y = \log_3 3x$

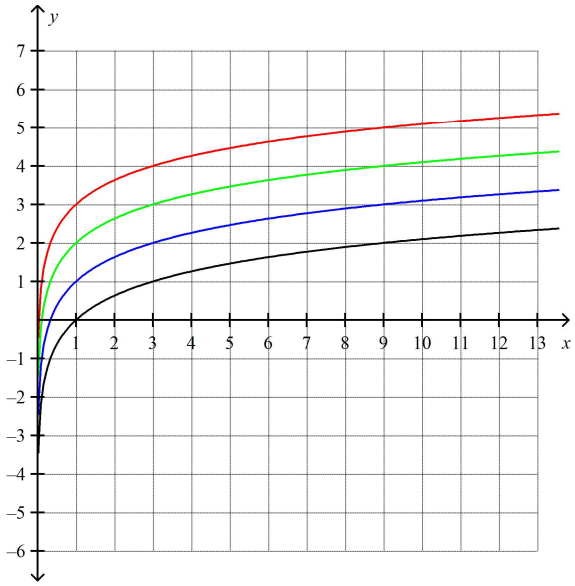
$y = \log_3 9x$

$y = \log_3 27x$

c) Explain the effect of the constant k in the function $y = \log_3 kx$.

ANS:

a), b)



c) The constant k will result in a vertical translation up by $\log_3 k$. Since k is of the form 3^n, where n is a whole number, the translation is up n units.

DIF: Average OBJ: Section 8.2 NAT: RF8
TOP: Transformations of Logarithmic Functions KEY: vertical translation

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17. Explain the steps used to solve the equation $\log_2 \sqrt{x^2 - 8x} = \log_2 3$.

ANS:

$\log_2 \sqrt{x^2 - 8x} = \log_2 3$

$\sqrt{x^2 - 8x} = 3$ Since the bases are the same, equate the logarithmic arguments.

$x^2 - 8x = 9$ Square both sides.

$x^2 - 8x - 9 = 0$ Express the quadratic equation in standard form.

$(x - 9)(x + 1) = 0$ Solve for x.

$x = 9$ or $x = -1$

Check the values for extraneous roots.

In this case, both values are possible.

DIF: Difficult OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: logarithmic equation | extraneous roots

18. Solve the equation $\log \sqrt[3]{x^2 + 48x} = \frac{2}{3}$.

ANS:

$\log \sqrt[3]{x^2 + 48x} = \frac{2}{3}$

$\log (x^2 + 48x)^{\frac{1}{3}} = \frac{2}{3}$

$\frac{1}{3} \log (x^2 + 48x) = \frac{2}{3}$

$\log (x^2 + 48x) = 2$

$\log (x^2 + 48x) = \log 100$

$x^2 + 48x = 100$

$x^2 + 48x - 100 = 0$

$(x + 50)(x - 2) = 0$

$x = -50$ or $x = 2$

Check the values for extraneous roots.

In this case, both values are possible and solve the equation, so they are both valid.

DIF: Difficult OBJ: Section 8.4 NAT: RF10
TOP: Logarithmic and Exponential Equations
KEY: logarithmic equation | extraneous roots

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19. For his dream car, Bruce invested $18 000 at 7.8% interest, compounded monthly, for 5 years. After the 5 years, he still did not have enough money. How much longer will he have to invest the money at 5% interest, compounded daily, to have a total of $35 000? Round to the nearest tenth of a year.

ANS:
For the first 5 years, the investment is compounded monthly for a total of 5×12, or 60, periods.

$$A = 18\,000(1 + 0.0065)^{60}$$

$$= 26\,552.12$$

Solve for the remaining time—compounded daily for n years is $365n$ periods.

$$35\,000 = 26\,552.12\left(1 + \frac{0.05}{365}\right)^{365n}$$

$$\frac{35\,000}{26\,552.12} = \left(\frac{365.05}{365}\right)^{365n}$$

$$\log\left(\frac{35\,000}{26\,552.12}\right) = 365n \log\left(\frac{365.05}{365}\right)$$

$$n \approx 5.53$$

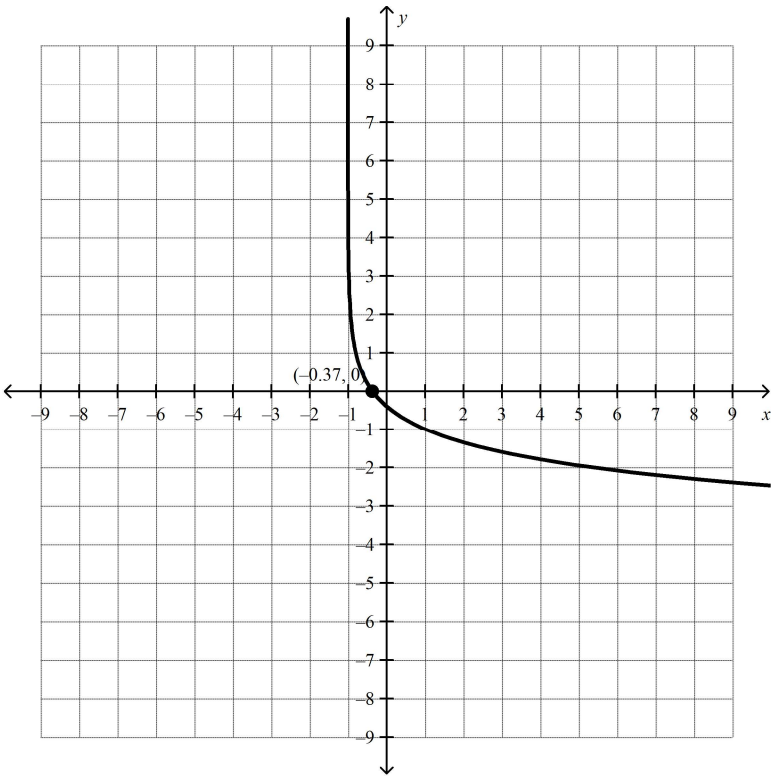
Bruce will need invest for approximately 5.5 more years.

DIF: Difficult OBJ: Section 8.4 NAT: RF10
TOP: Exponential and Logarithmic Equations
KEY: exponential function | logarithmic equation

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20. Sketch the graph of the function $f(x) = -2\log\left[\frac{1}{2}(x+1)\right] - 1$. Determine the x-intercept algebraically, to the nearest hundredth.

ANS:



Find the x-intercept by substituting $y = 0$ and solving for x.

$$0 = -2\log\left[\frac{1}{2}(x+1)\right] - 1$$

$$-\frac{1}{2} = \log\left[\frac{1}{2}(x+1)\right]$$

$$10^{-\frac{1}{2}} = \frac{1}{2}(x+1)$$

$$2(10^{-\frac{1}{2}}) - 1 = x$$

$$x \approx -0.37$$

The x-intercept is -0.37.

DIF: Average OBJ: Section 8.2 | Section 8.4 NAT: RF8 | RF10
TOP: Transformations of Logarithmic Functions | Logarithmic and Exponential Equations
KEY: transformation | logarithmic equation

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